CODEBREAKER TIPS

COUNTDOWN	A	B	C	D	HITS	ECHOES
7	0	0	0	1	1	0
6	0	2	2	3	2	0
5	0	2	4	4	1	1
4	0	5	2	5	2	0
3	0	6	2	7	2	1
2	0	7	2	8	2	0
1	0	9	2	6	4	0

This is an example of the last set of four numbers deciphered on the last guess.

7. Technically, it might be better to start with a guess of 0123, but you can get good results at the hard level by starting with 0001 and avoid some complications. Guessing three of one and one of another will give you better odds on interpreting the results correctly. In this case, odds are good that one of the zeros was correct.

6. On this turn I got two hits and figured it was likely a zero again and a 2 given that there are two 2s and only one 3. If I received echoes, I would have probably included a 3 in the next guess.

5. Here I thought either the zero or the 2 was an echo. I guessed the 2 and moved it to column C.

4. I figured the zero and 2 were now a lock for A and C.

3. Given the confidence about A and C, I tested two new numbers instead of one. The echo result meant either there's a 7 in column B or a 6 in column D.

2. I guessed that a 7 was in column B and was wrong. The 6 must have triggered the echo on the previous guess.

1. I knew 0926 would be correct. 9 was the only remaining possibility for column B because I had already tested a 6 and I knew there weren't additional zeros or 2s based on earlier guesses.

Multiplicity-weighted probing: I think an AAAB or AAB strategy is sound especially in the middle portion of the game. It's also good to test for positions while trying more numbers at the same time.

In another scenario, if I quickly determine more than one number. For example, if I've proven that the first three numbers are correct after guessing 0012, there a lot of remaining possibilities for the final number. Instead of guessing them one at a time (0013, 0014, etc), I will guess 0034, 0056, etc. If a new number gets a hit like the 6 in column D, I know the final result is 0016. But if a number gets an echo like the 5 in column C, I know the final result is 0015.

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Randomization: The launch code is completely random only half the time. The other half of the time, you'll encounter a special numerical pattern, such as the digits of an irrational number or a sequence with a mathematical significance. Sometimes the digits themselves will be randomly selected, but from a restricted set of numbers that share a common characteristic. Consider what Langford would select.

CODEBREAKER HISTORY

A game called Bulls & Cows emerged in the 1800s. It was played with paper and pencil. No one knows who invented it or where it came from. A computer version called MOO surfaced at Cambridge in the 1960s. A popular boardgame version called Mastermind was developed in 1970 by Mordecai Meirowitz, an Israeli postmaster and telecommunications expert.

With Bulls & Cows you were guessing numbers and repeat numbers were not allowed. With Mastermind you guess colors using pegs. In Mastermind there's a side column for smaller black and white pegs. Black pegs represent a correct guess, and white pegs represent a guess that's correct but in the wrong position. In Bulls & Cows, what the black peg represented was called a 'bull'. What the white peg represented was called a 'cow'.

In the late 1980s I wrote a version of Mastermind with Amiga Basic. It became Codebreaker in the early 2000s and the current HTML/JavaScript version was created in 2026. It uses numbers and repeat numbers are allowed. Bulls and cows are called hits and echoes, and non-random mathematical sequences were added. Computer-voice responses have been incorporated and the messages you hear are based on how well you are doing. The voice comments borrow from popular sci-fi and overall it has a War Games (1983) vibe.